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Unit 5 : Session 4 : SLO 2 | Symbolic Expressions and Algebraic Structures

I. Questions & Answers

1. Symbolic expression is a mathematical expression built from symbolic variables (declared with `sympy.symbols`) that is stored unevaluated, e.g. `x, y = symbols('x y');` `expr = x**2 + 2*y`.
2. Algebraic structure is a set equipped with one or more operations satisfying specific axioms, e.g. a group (closure, associativity, identity, inverse), a ring (addition + multiplication), or a field (a ring with multiplicative inverses).
3. Expanding $(x+1)^2$ in SymPy is `expand((x + 1)**2)` which gives `x**2 + 2*x + 1`.

II. Code

- `from sympy import symbols, expand`
- `x = symbols('x')`
- `print(expand((x + 1)**2))` # `x**2 + 2*x + 1`